Results outline

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June 3, 2026

Research questions

1. Does tree growth increase with longer growing seasons?
2. Did an earlier start and later end of season increase growth?
3. Does tree growth increase with longer growing seasons and is it consistent across species and provenances?
4. How did growth change across years, and how did it relate to climate?

Results

1. Summary basic information
   (a) Wildchrokie includes four deciduous tree species from four sites with leaf phenology data spaning three years (2018 to 2020) for a total of 239 leafout and 230 budset observations, yielding 227 complete growing seasons (GS) for 81 individuals.
   (b) Mean summer temperature were similar across years with 2019 being the coolest (22.38°C), 2020 the hottest (23.07°C), and 2018 in between (22.85°C)
   (c) The calendar and thermal seasons ranged from 90 days and 1465.1 GDD (*B. alleghaniensis* in 2018) to 173 days and 2513.6 GDD (*A. incana* in 2018)
   (d) The earliest leafout day of year was 22-Apr (*A. incana* in 2019) and the latest, 05-Jun (also *A. incana*, but in 2020). The earliest budset day of year was 13-Aug (*B. alleghaniensis* in 2018) and the latest, 24-Oct (*A. incana* in 2018)
   (e) No drought for the three years, except for in 2020, where a moderate summer drought ramped up to a severe drought in September.
   (f) Annual ring widths across years varied from 0.39 mm (*B. alleghaniensis* in 2020) to 13.55 mm (*B. populifolia* in 2018) (Figure S6).
2. Does tree growth increase with longer growing seasons?
   (a) Longer growing seasons increased growth for all species, but varied depending on the metric used to measure it.
   (b) Longer thermal season increased growth for *B. papyrifera* ($\beta = 0.44$, UI_{90} [0.18, 0.7]), *B. populifolia* ($\beta = 0.22$, UI_{90} [0.07, 0.36]), and *B. alleghaniensis* ($\beta = 0.13$, UI_{90} [0.01, 0.24]), but *A. incana* had a weak effect, but trended in the same direction ($\beta = 0.07$, UI_{90} [-0.04, 0.17]) (Figure 1a and e) (TableS1).
   (c) *A. incana* grew more with longer calendar seasons ($\beta = 0.05$, UI_{90} [0, 0.1]), but of the three species that responded to thermal season, only *B. papyrifera* also grew more with longer calendar seasons ($\beta = 0.18$, UI_{90} [0.05, 0.32]).
   (d) *B. alleghaniensis* ($\beta = 0.03$, UI_{90} [-0.02, 0.09]), and *B. populifolia* ($\beta = 0.05$, UI_{90} [-0.02, 0.13]) trended in the same direction (Figure 1b and f) (Table S1).
   (e) Results with basal area increment (BAI) for the GDD model were broadly similar to ring width (see Table S3)
   (f) On a scaled axis, both predictors were similarly predictive (FigureS7).
3. Did a later start and end of season increase growth?

(a) A longer calendar season increased growth for two species, but only one grew more with an earlier start of season (*A. incana*: β = -0.07, UI_{90} [-0.14, -0.01]) (Figure 1c and g) (Table S1).

(b) *B. alleghaniensis* which did not grow more with longer calendar season, grew less with an earlier start of season, unlike our prediction ($\beta = 0.1$, UI_{90} [0, 0.2]) (Figure 1c and g) (Table S1).

(c) Three species grew more with a later end of season (*B. alleghaniensis*: $\beta = 0.12$, UI_{90} [0.05, 0.18], *B. populifolia*: $\beta = 0.1$, UI_{90} [0.02, 0.17]), with *B. papyrifera* ($\beta = 0.21$, UI_{90} [0.09, 0.33]) being the only one of those three that also grew more with longer calendar season (Figure 1d and h) (Table S1).

(d) The estimates were unchanged when using the full (SOS $n = 239$ and $n = 230$) or restricted datasets (n =227) (Figure S1a).

4. How did tree growth change across years, provenances and tree ids?

   (a) We did not find any clear effect of year on growth (Figure S5) and there is no clear differences of growth looking across species (Figure S2 and Figure S3)

   (b) Growth did not change across any provenances (Figure 2) (Table S2), and the variation across sites ($\sigma =$ 0.18, UI_{90} [0.02, 0.5) and tree ids ($\sigma = 0.17$, UI_{90} [0.05, 0.27]) was similar

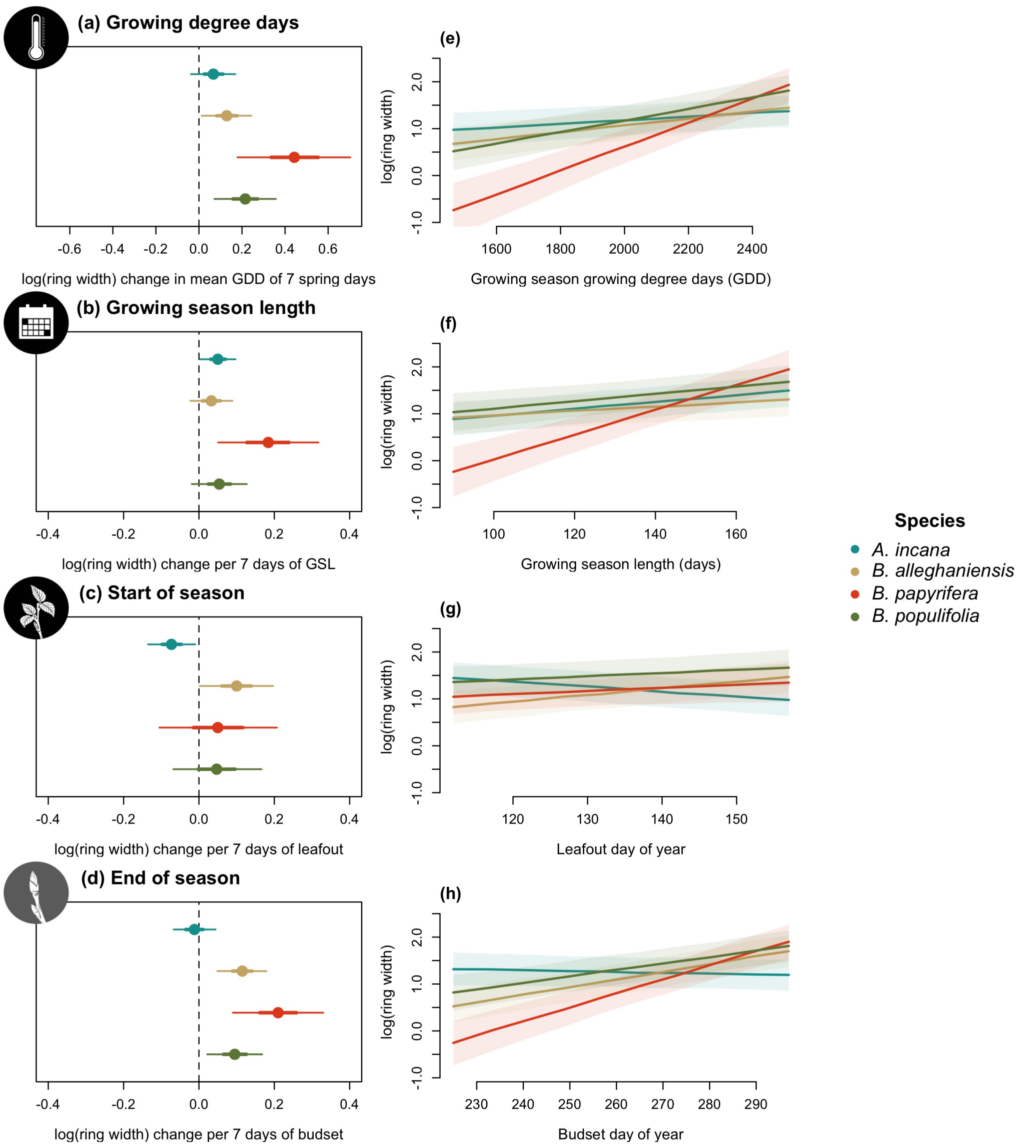

Figure 1: Model slope estimates representing ring width change as a function of four different predictors across four species. GDD represent the accumulated growing degree days between leafout and budset and the models estimates shows ring width change per 7 average spring days GDD (174.05) (a). GSL is the number of days between leafout and budset (b). SOS represent the start of season, which is leafout (c) and EOS, the end of season through budset (d). The dots represent the mean of each species and the lines, the 50% and 90% uncertainty intervals (UI).

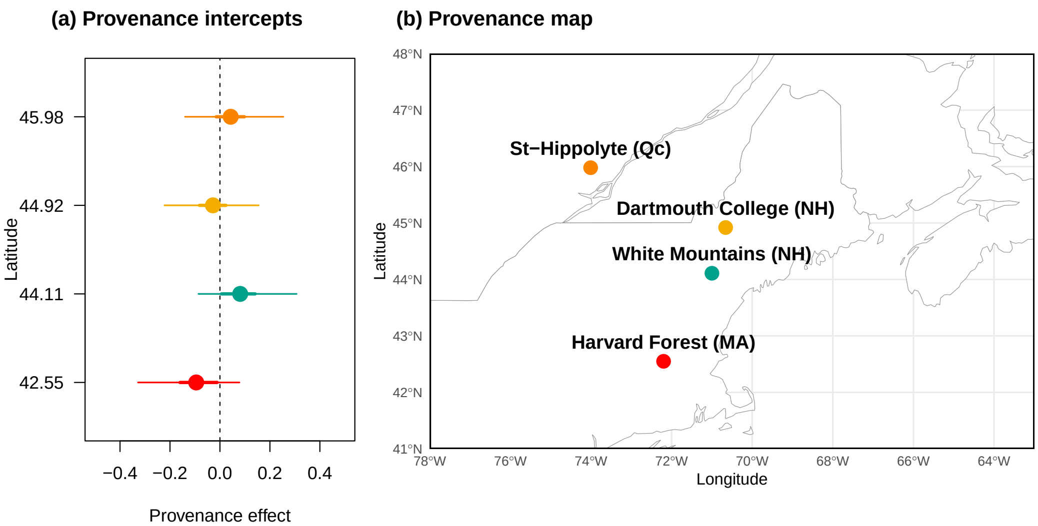

Figure 2: The effect of provenances on growth across a latitudinal gradient. (a) Model intercept parameters for each site with growing degree days (GDD) as the predictor (effects were similar for the other predictors, see Table S2). The dots represent the mean of each provenance and the thick lines, the 50% and 90% uncertainty intervals (UI). The sites are color-coded from (b), the map showing their locations.

Supplemental results

Table S1: Species level slope parameters (mean and 90% uncertainty intervals, in square brackets) across the four different predictors for Wildchrokie. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

Species	GDD	GSL	SOS	EOS
A. incana	0.07 [-0.04, 0.17]	0.05 [0.00, 0.10]	-0.07 [-0.14, -0.01]	-0.01 [-0.07, 0.04]
B. alleghaniensis	0.13 [0.01, 0.24]	0.03 [-0.02, 0.09]	0.10 [0.00, 0.20]	0.12 [0.05, 0.18]
B. papyrifera	0.44 [0.18, 0.70]	0.18 [0.05, 0.32]	0.05 [-0.10, 0.21]	0.21 [0.09, 0.33]
B. populifolia	0.21 [0.07, 0.36]	0.05 [-0.02, 0.13]	0.05 [-0.07, 0.17]	0.10 [0.02, 0.17]

Table S2: Site level intercept parameters (mean and 90% uncertainty intervals, in square brackets) across the four different predictors for Wildchrokie. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

Site	GDD	GSL	SOS	EOS	Lon.
Harvard Forest (MA)	-0.10 [-0.33, 0.08]	-0.10 [-0.34, 0.08]	-0.05 [-0.26, 0.11]	-0.09 [-0.33, 0.08]	-72.19
White Mountains (NH)	0.08 [-0.09, 0.31]	0.07 [-0.10, 0.29]	0.08 [-0.07, 0.29]	0.08 [-0.10, 0.30]	-71.23
Dartmouth College (NH)	-0.03 [-0.22, 0.15]	-0.05 [-0.26, 0.13]	-0.07 [-0.27, 0.09]	-0.04 [-0.24, 0.15]	-71.15
St-Hippolyte (Qc)	0.04 [-0.14, 0.25]	0.03 [-0.16, 0.24]	0.02 [-0.15, 0.21]	0.05 [-0.14, 0.27]	-74.03

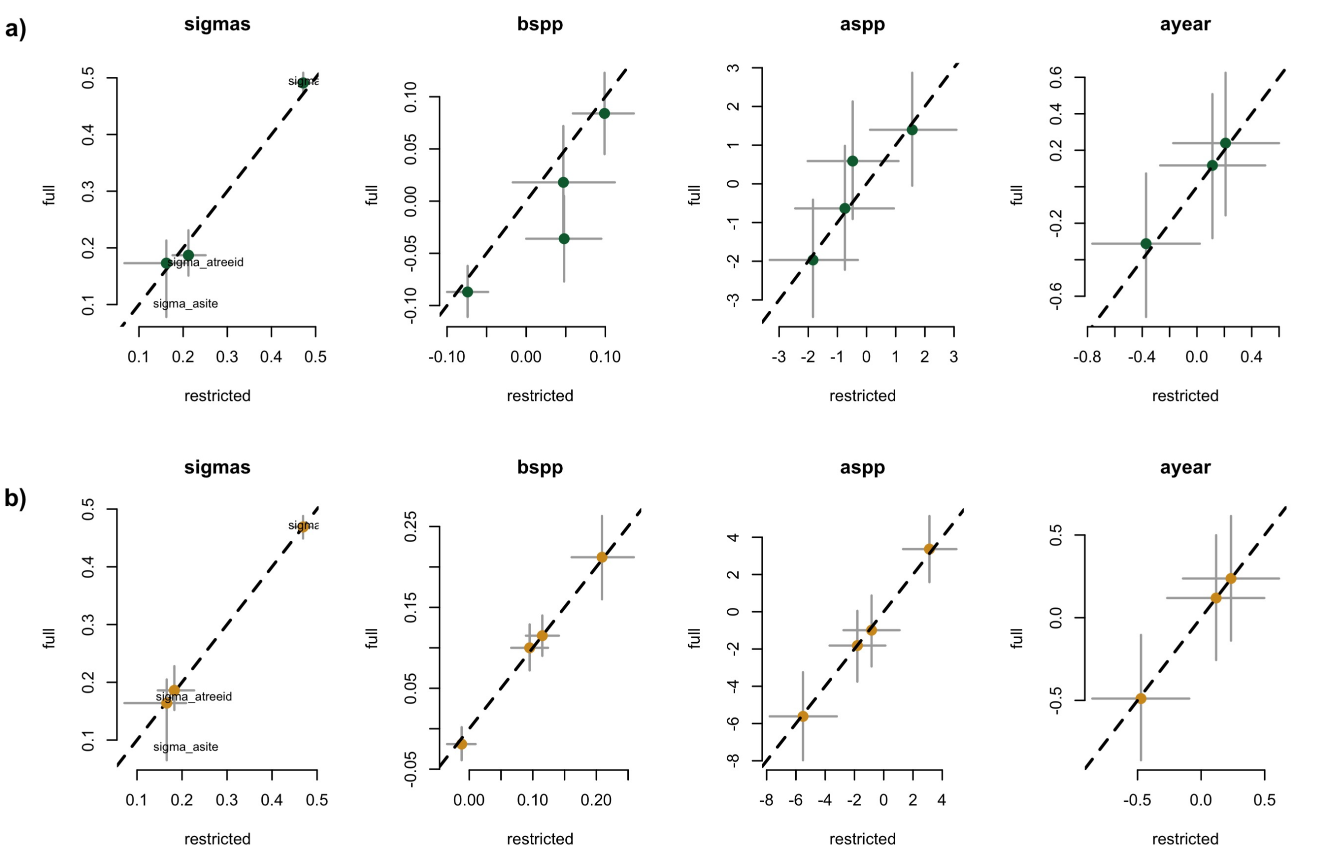

Figure S1: Full vs most restricted rows of data for the model with SOS (a) EOS (b) as the predictor for Wildchrokie. Each pannel represent the different parameters used to fit the models.

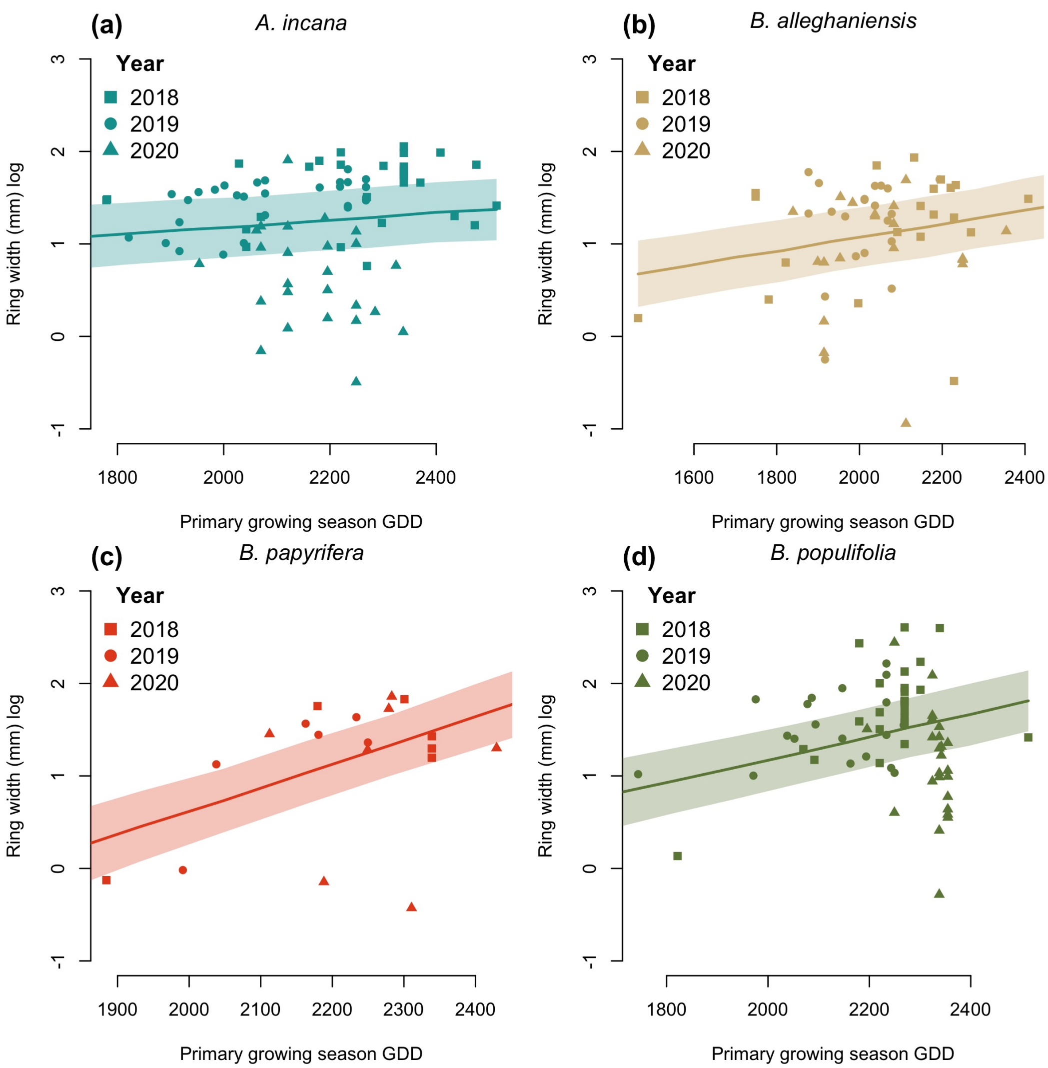

Figure S2: Ring width trends in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

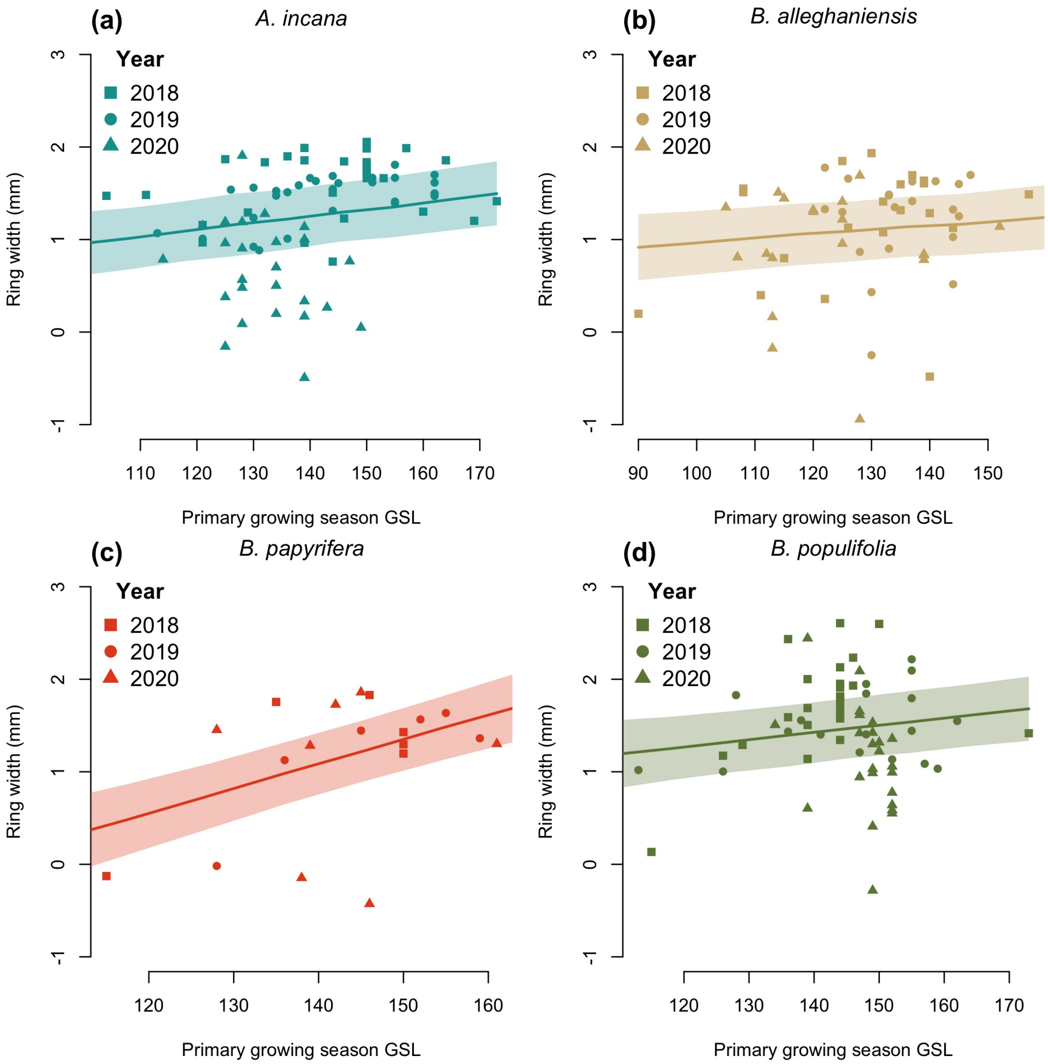

Figure S3: Ring width trends in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

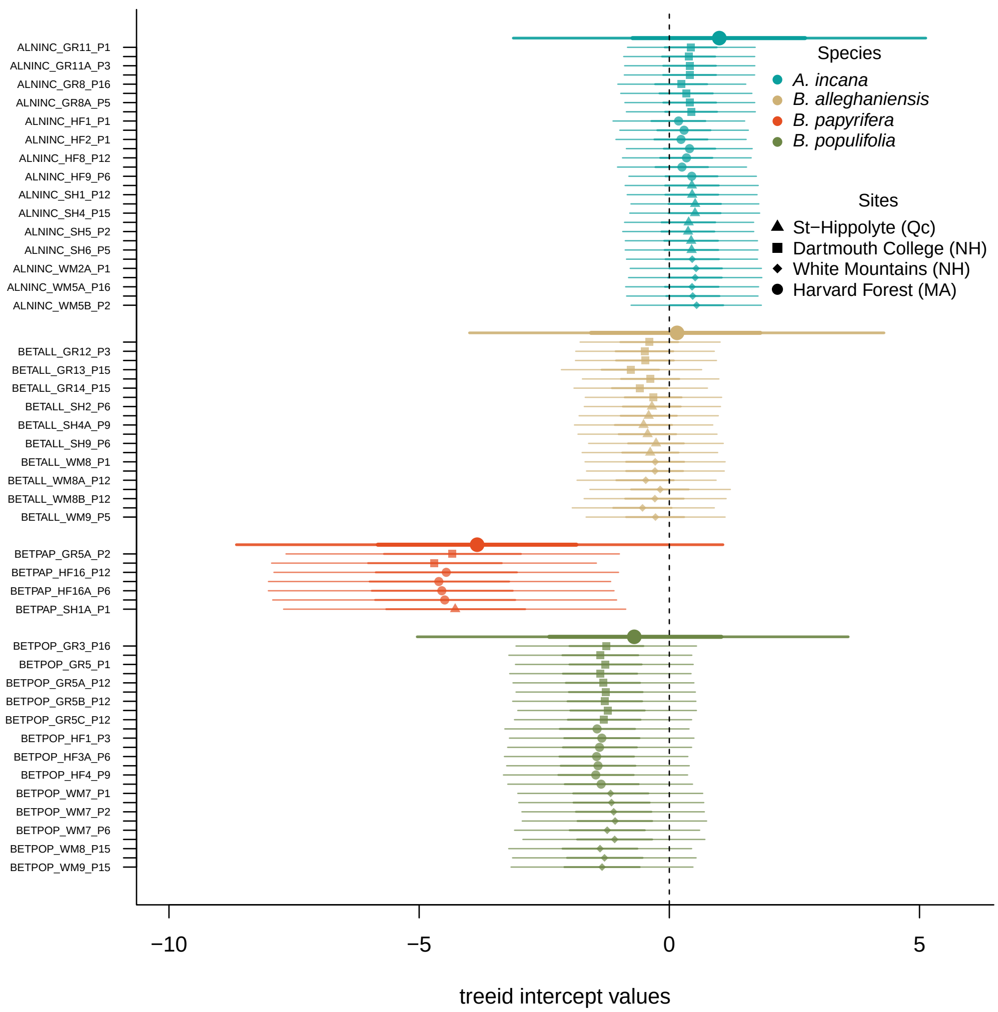

Figure S4: Intercept estimates for each group of the GDD model. The different tree id values represent the combined intercepts of tree id, provenance, species and averaged year intercepts. The bigger dots and lines represent the species intercepts which are the species level estimates averaged across all of their tree ids. The larger dots and lines represent aggregated average across these values. The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

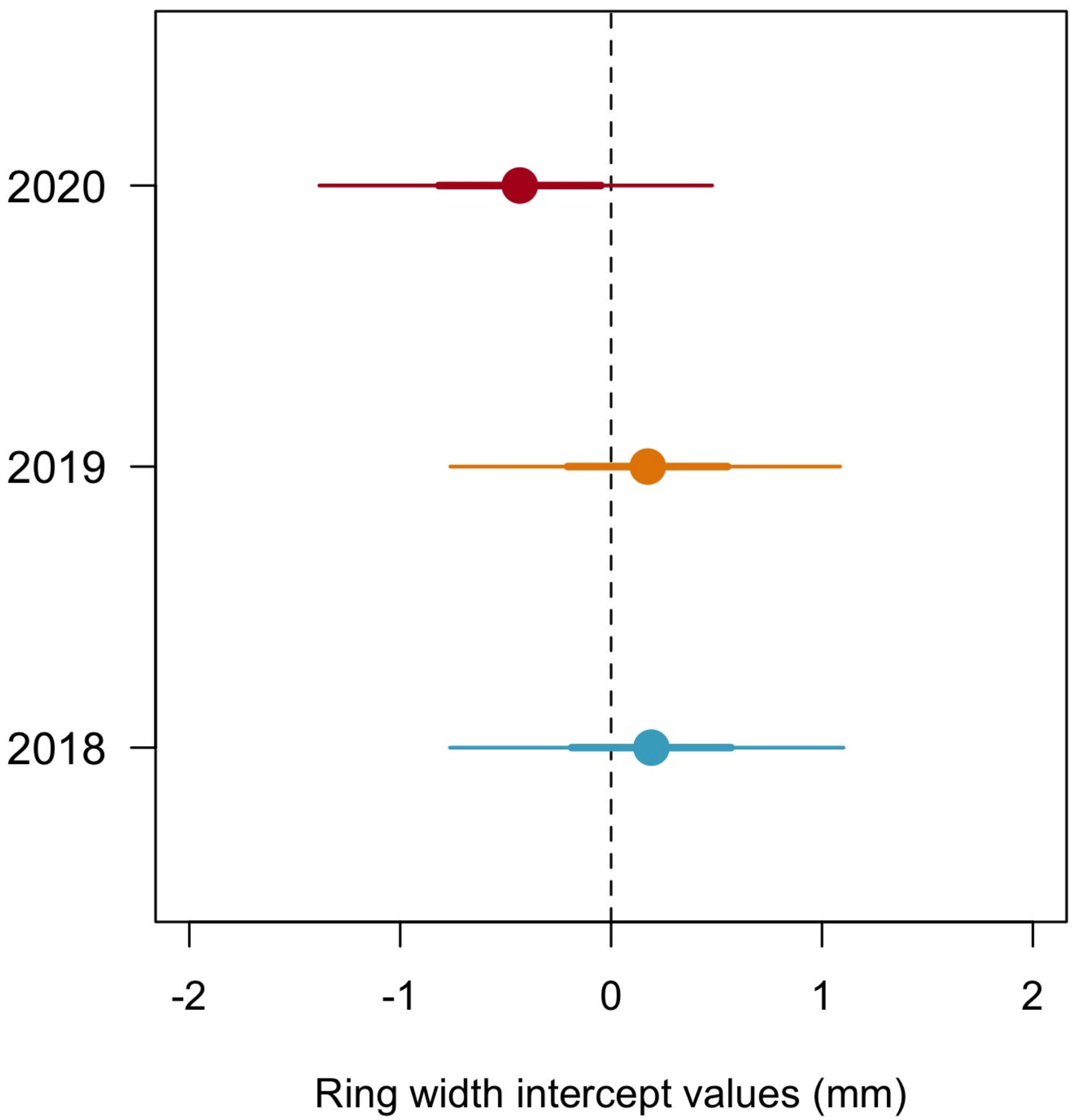

Figure S5: Model estimates for the year intercepts with the dots depicting the mean and the lines, the 50% and 90% uncertainty intervals (UI) and (b) using box plots of the empirical data. Though the trend is weak, 2020 is systematically the year with the lowest growth shown both as the mean of the model intercepts (a) and as the median of the raw data, for each species (b). The growth of 2018 and 2019 is similar.

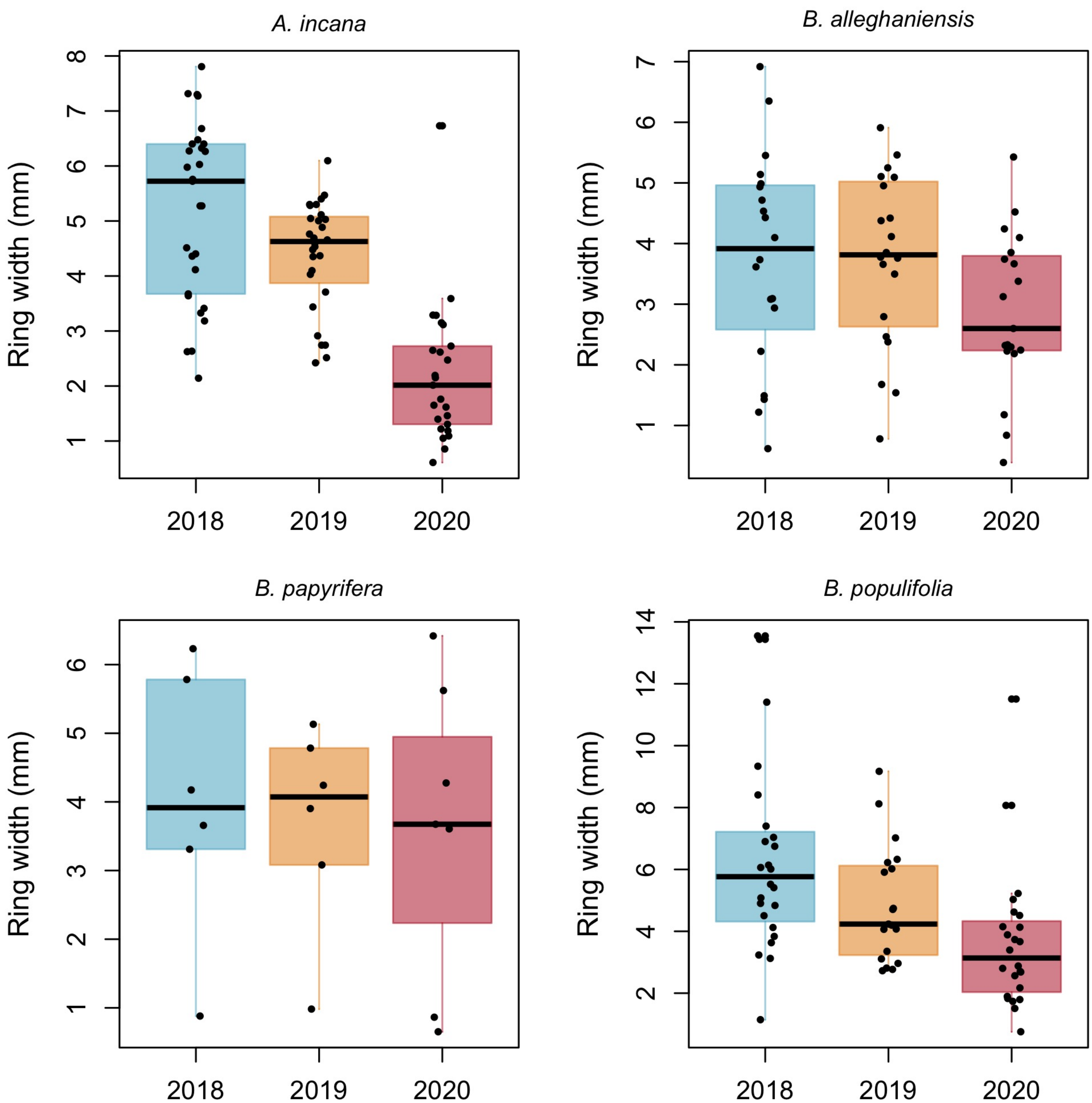

Figure S6: Boxplot representing the empirical data of our ring widths (mm) facetted by species. The dots represent the raw data, the line the mean and the box the IQR.

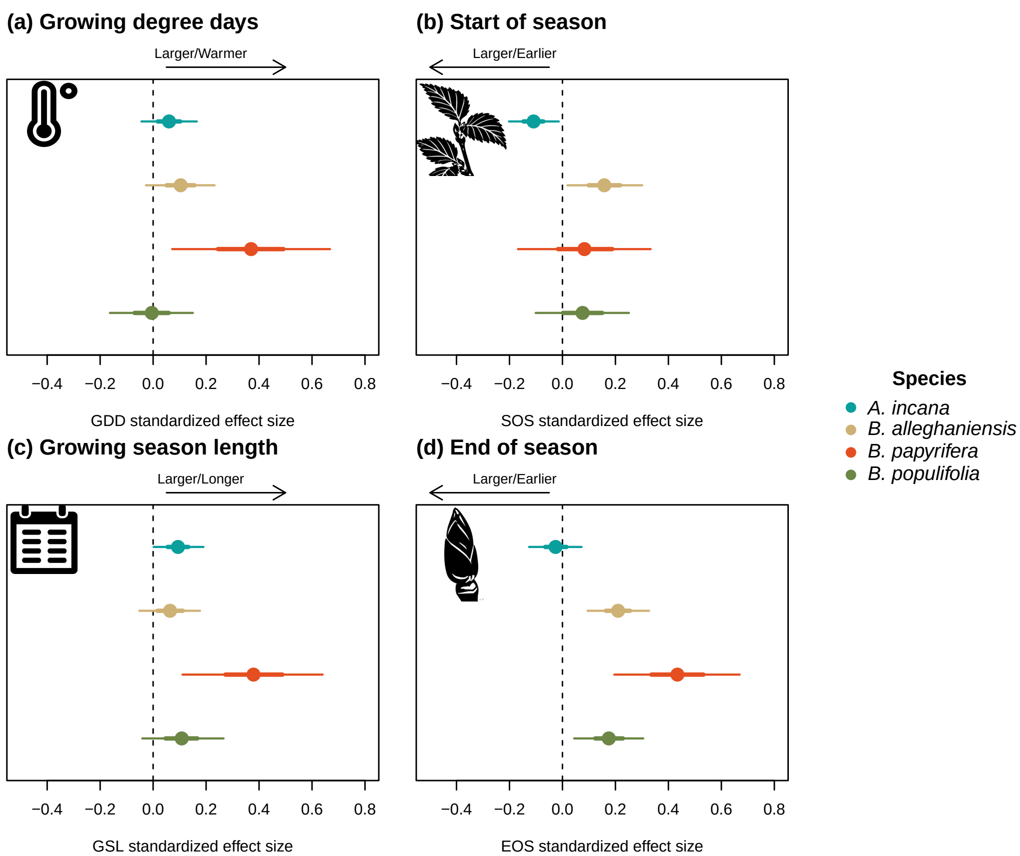

Figure S7: Standardized (z-scored) slope estimates for Wildchrokie. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI.

Table S3: Posterior means and 90% uncertainty intervals for all model parameters, when using basal area increment (BAI) as the response variable

Parameter	Estimate
$\sigma_{\alpha_{tree}}$	0.52 [0.41, 0.64]
$\sigma_{\alpha_{site}}$	0.27 [0.03, 0.73]
σ_y	0.63 [0.57, 0.70]
$\beta_{spp_{A.incana}}$	0.11 [-0.03, 0.25]
$\beta_{spp_{B.alleghaniensis}}$	0.20 [0.02, 0.36]
$\beta_{spp_{B.papyrifera}}$	0.92 [0.49, 1.34]
$\beta_{spp_{B.populifolia}}$	0.41 [0.20, 0.61]
$\alpha_{site_{DartmouthCollege(NH)}}$	-0.11 [-0.47, 0.17]
$\alpha_{site_{HarvardForest(MA)}}$	-0.09 [-0.45, 0.19]
$\alpha_{site_{St-Hippolyte(Qc)}}$	0.06 [-0.23, 0.42]
$\alpha_{site_{WhiteMountains(NH)}}$	0.15 [-0.11, 0.52]
$\alpha_{year_{2018}}$	-0.36 [-1.32, 0.61]
$\alpha_{year_{2019}}$	0.33 [-0.65, 1.32]
$\alpha_{year_{2020}}$	-0.04 [-1.01, 0.94]
$\alpha_{spp_{A.incana}}$	3.23 [-2.25, 8.55]
$\alpha_{spp_{B.alleghaniensis}}$	2.09 [-3.40, 7.54]
$\alpha_{spp_{B.papyrifera}}$	-7.59 [-14.84, -0.29]
$\alpha_{spp_{B.populifolia}}$	-0.07 [-5.98, 5.57]